



EXPENSE DASHBOARD PRO

MOBILE APPLICATION DEVELOPMENT



GROUP MEMBERS:

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Project Title: Expense Dashboard Pro

Course: Mobile Application Development (MAD)

Platform: Android Native

Language: Kotlin **UI Framework:** Jetpack Compose

1. Project Overview

The **Expense Dashboard Pro** is a native Android application designed to serve as a comprehensive, real-world financial tracking tool for university students. Moving beyond standard basic inputs, this application focuses on a premium User Experience (UX) and dynamic state management, allowing users to log their daily expenses across specific categories (like Food, Transport, and University). The core objective was to build a functional, bug-free application that heavily utilizes modern Jetpack Compose architecture to present aggregated data visually.

2. Core Features

- **Dynamic Visual Charts:** The main screen features a visual breakdown of expenses using dynamically calculated, colored horizontal bars.
- **Smart Budget Warnings:** Users can define an optional custom monthly budget; the app calculates spending and dynamically turns the total text red while issuing a warning if the limit is exceeded.
- **Complete CRUD Functionality:** Users can create new expense records and delete individual transactions from their history to fix mistakes.
- **Persistent Storage:** The application utilizes local storage methods to permanently save transaction history and budget settings, ensuring data persists even after the app is fully closed.
- **Automated Timestamps:** The system automatically captures the exact date and time the moment a user logs an expense.
- **Continuous UI Scrolling:** The entire dashboard is wrapped in a full-page scrollable layout, allowing the interface to adapt seamlessly as the transaction list grows.

3. Technical Architecture & MAD Concepts Demonstrated

Feature / Concept	Technical Implementation	Why it was used
Modern UI Layouts	Jetpack Compose	It is the modern standard for Android UI, allowing for faster, cleaner code written directly in Kotlin.

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State Management	remember & mutableStateOf	Ensures the UI updates instantly in real-time when expenses are added or deleted without freezing the application.
Smooth Animations	animateFloatAsState	Allows the visual chart bars to physically grow and shrink with a smooth animation whenever the dataset changes.
Data Persistence	SharedPreferences	A lightweight solution to handle local data saving, successfully storing both the categorized expense strings and the user's custom budget.
List Rendering	LazyColumn	Replaces traditional XML RecyclerView to efficiently render an infinite, scrollable list of recent transactions.

4. User Experience (UX) Polish

Great attention was paid to the usability of the application. Instead of vague, generic input fields, the app prompts users with context-specific examples (e.g., logging "Bus fare" under Transport or a "Semester fee challan" under University) to accurately reflect student life. Furthermore, forcing a hardcoded budget was avoided in favor of a flexible, user-defined input, treating the user with a highly professional design standard.

5. Conclusion

The Expense Dashboard Pro successfully demonstrates mastery over modern Android development. By prioritizing a clean architecture and avoiding scope creep, the final product is a highly polished, interactive, and functional application that fully satisfies the rigorous requirements of the Mobile Application Development syllabus.